SeeBFM: Behavior Foundation Model That Sees

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https://capfab.github.io/seebfm-media/

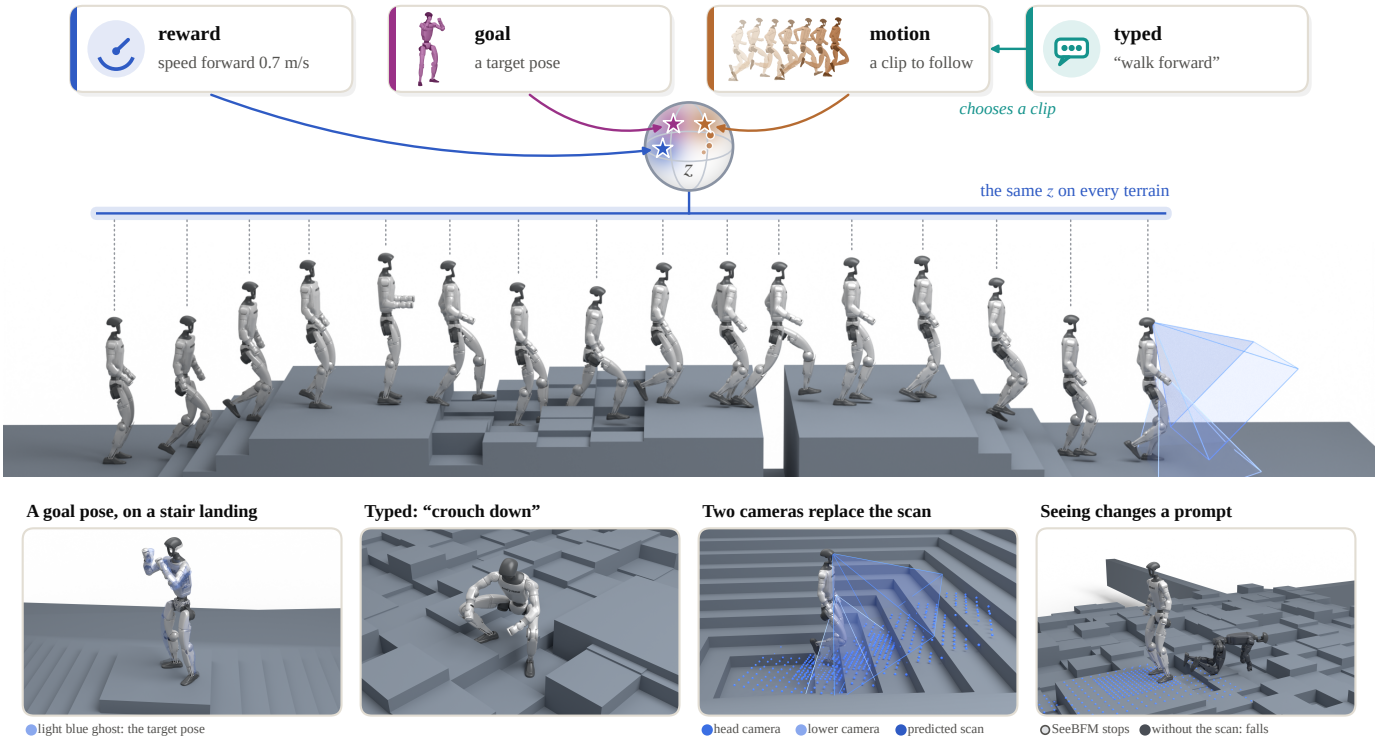


Fig. 1. Rewards, goals and motions become latent prompts; typed commands select motion prompts. **SeeBFM** uses terrain perception to execute them.

***Abstract*—Behavior foundation models turn a reward, a goal pose or a motion into a prompt for a single pretrained humanoid policy, but models such as BFM-Zero cannot see the ground. Perception should change how the robot carries out a prompt and leave what the prompt asks for unchanged: *prompt the what, perceive the how*. Since a learned network computes each prompt, where perception enters decides whether this holds. SeeBFM gives BFM-Zero a height scan during continued pretraining that every network reads except the backward network that encodes prompts, so a reward asks for the same body states on every terrain. The new pathways start at zero, so before training SeeBFM's networks compute exactly what BFM-Zero's do on the same inputs. In simulation, with the same six locomotion prompts, SeeBFM crosses stairs going up in more than half of its attempts and boxes in two of five, where BFM-Zero crosses neither, and its forward prompt slows on stairs and lifts the swing foot over each step. Crossing drops sharply when the scan is made level or omitted during training. Measured prompt by prompt, perception also changes how prompts are executed: walking forward and turning carry over to terrain, while walking backward nearly stops and walking sideways slows to less than half its flat-ground speed. On stairs going up, SeeBFM crosses with its prompts alone, while a concurrent perceptive model relies on an additional heading and velocity command. A depth student replaces the scan with predictions from two cameras without retraining the policy, and typed commands select motion prompts for the same policy.**

I. Introduction

Behavior foundation models are pretrained once, without a task, and then prompted. A reward function, a goal pose or a motion to imitate becomes a latent z, and a single policy conditioned on z produces the behavior with no further training [28, 15]. For a humanoid this is an attractive interface, since one policy serves every task its prompts can express. However, these released models read proprioception and the latent only, without observations of the ground. On stairs going up with 0.10 to 0.15 m risers the released BFM-Zero for the Unitree G1 crossed none of 226 attempts in our evaluation. The authors of Meta Motivo list perception among the model's limits and expect that adding it “might necessitate test-time planning or fast online adaptation” [28].

Perception has reached humanoid control mainly through

motion trackers, whose command is a reference motion and whose policy decides how to follow it on the terrain at hand [36, 10, 12]. A promptable model raises an additional question. It computes prompts with its learned backward network, so changing the representation can also change how a reward is encoded. Where should perception enter so that a reward keeps asking for the same behavior, and what does each prompt do once the policy can see?

The forward-backward factorization behind BFM-Zero suggests an answer. It splits the robot's future into a backward network B, which embeds the states a behavior should reach, and a forward network F, which predicts how an action leads there. A reward is encoded as a reward-weighted average of B over reference states. A reward on the body's state favors the same states on every terrain, and the terrain decides how they can be reached. SeeBFM therefore gives a height scan to the policy, F, the critics and the discriminator, and leaves B reading the body alone. In the terms of a navigation system, the prompt is the destination and the scan is the road: the destination stays fixed while the route adapts to the ground. We call this principle ***prompt the what, perceive the how***. The interface stays the same, but continued training can still change the latent a reward is encoded as and what that latent makes the robot do, and we evaluate both.

We build SeeBFM by continuing BFM-Zero's training from its released weights on 379 h of motion, 78 h of it on stairs, boxes and uneven ground. Each new input enters through a pathway that starts at zero, preserving BFM-Zero's network outputs for the same base inputs at initialization. Holding these pathways at zero gives a control, the model trained without the scan, that learns from the same motion corpus without height-scan input. The discriminator reads the scan too, so that it can assess motion in its terrain context.

Our contributions are:

- a way to add perception to a pretrained behavior foundation model while retaining its prompt interface: the scan enters every network except the backward network B;
- SeeBFM, which crosses stairs and boxes in simulation when prompted with the same reward prompts under which BFM-Zero falls;
- the finding that perception changes prompt execution: walking forward and turning carry over to stairs and boxes, while walking backward nearly stops and walking sideways slows by more than half;
- a depth student that replaces the scan with two cameras without retraining the policy, and typed commands that reach the same skill.

II. Related work

Behavior foundation models. Forward-backward representations [30] approximate the successor measure and infer a policy from a reward-weighted mean of B over reference states. FB-CPR [28] regularizes the policy toward motion capture with a discriminator, and BFM-Zero [15] scales the approach to a humanoid and evaluates prompts over its replay buffer. Other humanoid behavior models distill a motion tracker into a conditional latent space [41] or span several robots from proprioception and sparse tracking signals [43], and none of them reads the terrain. Sikchi et al. [25] adapt a trained model's latent to find better policies than its inference does. Bobrin et al. [1] adapt it to unseen dynamics and find that conditioning B on an inferred context smooths the value function and degrades performance, so they keep B shared across contexts; we measure the corresponding variant on terrain. We study this separation for terrain perception and test centering B when the inference states differ from the training distribution.

Perceptive locomotion. Dedicated perceptive policies reach hard terrain from height maps, depth or point clouds. PRIOR [6] estimates a height map from depth with a recurrent network and follows a parametric gait prior, TAGA [13] attends to the informative regions of a height scan, Hiking in the Wild [48] runs parkour end to end from depth on a G1, CReF [7] fuses depth and proprioception through cross-modal attention, and UniPoint [14] fuses LiDAR and two depth cameras at the level of points. TACT-ful [16] turns a terrain cost of flatness, steepness and height feasibility into both foothold targets and a reward for a policy trained from depth, and Ly et al. [17] route five experts by a terrain classifier over a sensed elevation grid and place each foothold with a model-based planner. These policies are trained for locomotion alone and have no prompt interface to preserve.

Perceptive motion tracking. Perceptive BFM [36] lets a height map enter a pretrained tracker through zero-initialized gates on its intent latent and action while the tracker keeps training at a reduced rate, and ViBe [10] trains zero-initialized low-rank adapters inside a frozen tracker. PGMT [12] first learns a general tracking prior and then adds terrain perception through glimpses selected by the current motion, and GigaBrain-WBC [4] learns a behavior world model whose dynamics depend on the environment, for tracking across terrain. Generator-tracker systems reach several perceptive skills by letting a terrain-aware motion generator propose motion for a tracker to follow [20, 46]. Scene-aligned human motion lets a planner and a tracker cross clutter [18], and immersive human demonstrations train scene-aware locomotion through it [33, 32]. These systems use motion references or learned generators, and none infers its command from a reward.

Adding an input to a trained model. ControlNet [44] adds a condition to a frozen image model through layers that start at zero. Perceptive BFM and ViBe use the same device, and so does SeeBFM, here inside the pretraining objective of a promptable model with every network training. Zero-initialized additions match the base function at initialization; modality-gated adapters can preserve the original mapping when the added modality is absent [29]. Residual policies learn a correction on top of a fixed controller [26, 9], latent steering acts on a frozen generative policy's input noise, alone or with a residual action [31, 27], and on a quadruped a terrain-aware policy steers a frozen flat-ground prior through its latent commands and joint residuals [45]. A KL penalty to the pretrained model is a common way to keep fine-tuning

close to it [49]; SeeBFM preserves the initial network function for fixed inputs and lets every network keep training.

Style priors and data on terrain. T-GMP [5] conditions an adversarial motion prior on a local height map, so that terrain motion is judged against terrain motion, and SeeBFM's discriminator reads the scan for the same reason. Zeng et al. [42] find that a behavior foundation model gains more from mixing sources of motion than from more of one source, which favors adding terrain datasets over more flat motion.

Language. Shvetsov et al. [24] align sequences of a frozen model's latents with text and decode them back into latents for long composite motions, and TANGO [11] trains a whole-body vision-language-action model for navigation in clutter from instructions.

Concurrent work. P-BFM [35], developed concurrently with SeeBFM, trains a perceptive forward-backward model from scratch with the same division of inputs: B reads the body, while the forward network, the critics and an actor reading depth images see the terrain. Because latent-only control drifted in heading and stepped in place before obstacles, P-BFM adds a heading and velocity command beside the latent. It learns from 40 flat-ground motions with a discriminator that does not see the terrain, reports tracking and recovery, and shows a real G1 on stairs, slopes and gaps. SeeBFM continues BFM-Zero's training on 78 h of terrain motion with a discriminator that reads the scan, measures how each prompt behaves on terrain against controls that take the scan away, and is compared with the authors' P-BFM policy on matched terrain dimensions in their respective simulators.

III. Method

We extend BFM-Zero through three design choices: what each network sees, how we initialize the new inputs, and what the style prior compares against (Fig. 2). A fourth change concerns how prompts are read out after training. We take them in turn, then describe the depth student and typed commands, which leave the trained policy as it is.

A. Prompts in a forward-backward model

BFM-Zero [15] learns a forward network $F(s, a, z)$, a backward network $B(s) \in \mathbb{R}^{256}$ and a policy $\pi(a \mid s, z)$, with z on the sphere of radius 16. The two networks have distinct roles: B defines the features used to express a task, and F predicts their discounted occupancy after an action. A reward prompt therefore asks for body states, and the policy learns how to reach them. Here s denotes body-state features. We suppress observation histories in the network notation; the policy uses proprioceptive observations, while training can use privileged body state. The networks approximate the successor measure of π_z, the expected discounted occupancy of future body states s_{t+1} with weights γ^t, $t \geq 0$,

$$M^{\pi_z}(s, a, \mathrm{d}s') \approx F(s, a, z)^\top B(s')\, \rho(\mathrm{d}s'), \tag{1}$$

where ρ is the training reference distribution over body states [30, 28]. For a reward r, define the unnormalized embedding $u_r = \mathbb{E}_{s\sim\rho}[r(s)B(s)]$. For a fixed policy π_z, the factorization gives $Q_r^{\pi_z}(s, a) \approx F(s, a, z)^\top u_r$. The basic reward prompt is $z_r = \mathrm{proj}(u_r)$, where $\mathrm{proj}(u) = 16u/\|u\|_2$ for $u \neq 0$. The value estimate above still uses the unnormalized u_r. The actor is trained to increase $F^\top z$, together with the regularizers below, so it approximates, without guaranteeing, the best policy for every reward.

B. Terrain perception

Every reward we prompt with is a function of the robot's body state. A height scan σ describes the terrain that affects how these states can be reached, without changing the reward function. We therefore condition only the forward factor on it,

$$M^{\pi_z}(s, \sigma, a, \mathrm{d}s') \approx F(s, \sigma, a, z)^\top B(s')\, \rho(\mathrm{d}s'), \tag{2}$$

and give the scan to the policy, F, the critics and the discriminator. This measure describes future body states, with future terrain observations marginalized out; it does not model their joint occupancy. For a fixed trained B and inference sample set, a body-state reward gives the same prompt on every terrain. Its estimated value still depends on the scan through F. This separates task specification from terrain-dependent execution without assuming that a local scan fully observes the environment. Giving B the scan would require terrain observations when encoding target states. We test this alternative in the tracking evaluation. B's input changes in one respect: the root height is measured above the ground under the robot, which on flat ground is the released input. F and B are trained with the temporal-difference loss of Touati and Ollivier [30], the scan on the forward side,

$$\begin{aligned}\mathcal{L}_{FB} = \mathbb{E}\Big[\tfrac{1}{2}\big(F(s, \sigma, a, z)^\top B(s^+) \\ - \gamma\, \widehat{F}(s', \sigma', a', z)^\top \widehat{B}(s^+)\big)^2 \\ - F(s, \sigma, a, z)^\top B(s')\Big] + \lambda_\perp\, \mathcal{L}_\perp(B),\end{aligned} \tag{3}$$

where $(s, \sigma, a, s', \sigma')$ is a transition from the replay buffer, s^+ a state drawn from it independently, $a' \sim \pi(\cdot \mid s', \sigma', z)$, the hats mark target networks, and $\mathcal{L}_\perp = \mathbb{E}\big[\frac{1}{2}(B(s)^\top B(s^+))^2\big] - \mathbb{E}\big[B(s)^\top B(s)\big]$ encourages $\mathbb{E}_\rho[B(s)B(s)^\top] \approx I_{256}$, with $\lambda_\perp = 100$. It constrains B's second moment and leaves its mean free; centering (Eq. 7) returns to this.

C. Starting from the released weights

A promptable model is valuable for the behaviors its prompts already produce, so SeeBFM grows from the released weights. A scan encoder e enters the actor, F, both critics and the discriminator after the first block of each network,

$$h = \mathrm{Block}_0(x) + W\, e(\sigma), \tag{4}$$

where x is each network's original input and $W = 0$ at initialization. The pathway joins after the first block because BFM-Zero's residual networks open with a layer normalization over the whole input. Appending a scan there would change the normalization's mean and variance even if its input weights were zero.

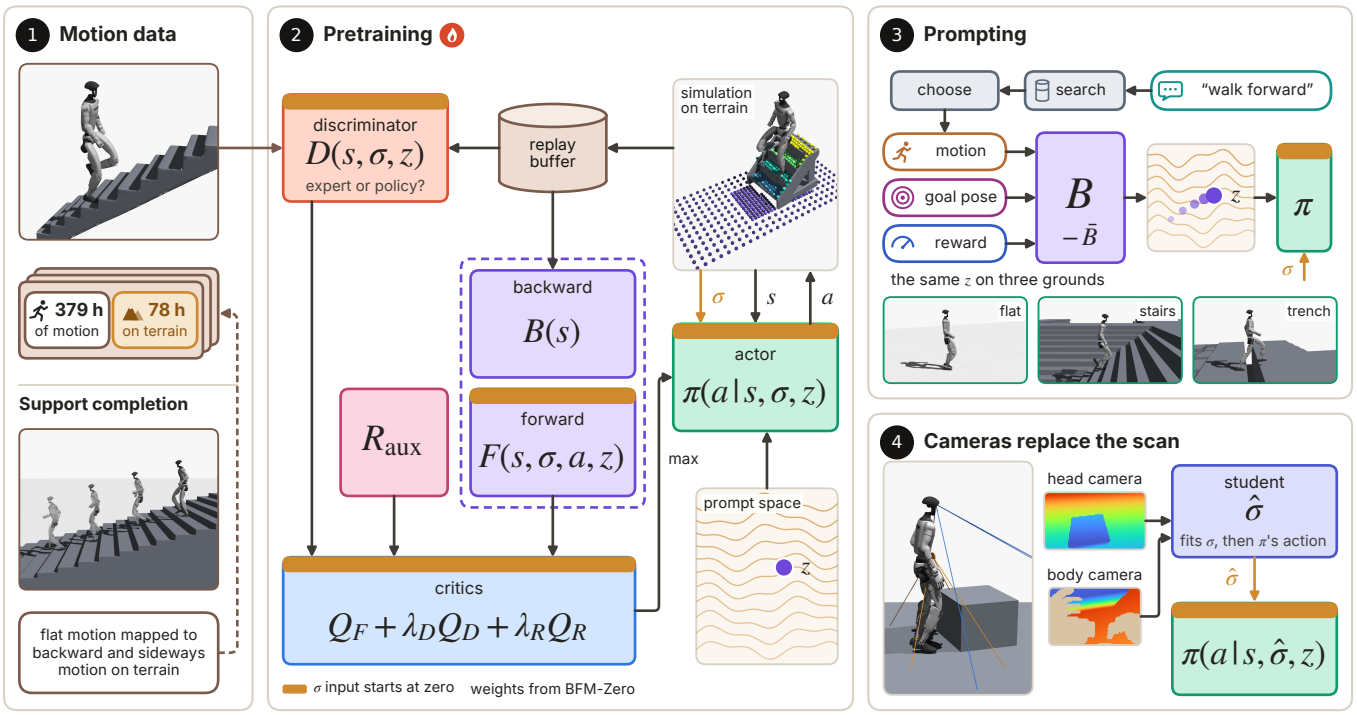


Fig. 2. **SeeBFM pipeline**. (1) Motion on terrain. (2) Pretraining continues from BFM-Zero. (3) Rewards, goals and motions become latent prompts z; typed commands select a motion and its prompt sequence. (4) A depth student predicts σ from two cameras.

Proposition 1: At initialization, with the released weights, identical base inputs and preprocessing, and $W = 0$ in every pathway, the actor, F, both critics and the discriminator equal the released networks for every scan σ. For a single example, the gradient of a loss $\mathcal{L}$ with respect to a pathway's W is $(\partial \mathcal{L}/\partial h)\, e(\sigma)^\top$, so the pathway receives gradient from the first update.

Both parts follow from (4): at $W = 0$, h equals the first block's output, and the chain rule gives the stated gradient. This holds at initialization only: continued training changes behavior, and on uneven terrain so does the new root-height input. The same construction yields a clean control: holding every W at zero trains the same networks on the same data and seeds, with no height-scan input reaching any of them.

D. A style prior that sees the ground

BFM-Zero keeps its policy close to motion capture with the discriminator of FB-CPR [28]. Terrain changes which motions are appropriate, so we also condition the discriminator on the scan. We cast the scan at the expert's pose for expert states and at the robot's pose for policy states. An expert window τ is encoded as $z_\tau = \text{proj}(|\tau|^{-1} \sum_{s \in \tau} B(s))$, and the discriminator is trained with a logistic loss and a gradient penalty,

$$\begin{aligned} \mathcal{L}_D = &- \mathbb{E}_{\mathcal{M}}\big[\log D(s, \sigma, z_\tau)\big] \\ &- \mathbb{E}_{\pi}\big[\log\big(1 - D(s, \sigma, z)\big)\big] + \lambda_{\text{gp}}\, \mathcal{L}_{\text{gp}}, \end{aligned} \tag{5}$$

over expert windows from the motion data $\mathcal{M}$ and the policy's own states, with $\lambda_{\text{gp}} = 10$. The log-ratio $\log D - \log(1 - D)$ and BFM-Zero's auxiliary penalties have critics Q_D and Q_R, trained by temporal differences, and the actor maximizes

$$\mathbb{E}_{a \sim \pi(\cdot \mid s, \sigma, z)}\big[F(s, \sigma, a, z)^\top z + c_Q\, (\lambda_D Q_D + \lambda_R Q_R)\big], \tag{6}$$

where c_Q is the batch mean of $|F^\top z|$, treated as constant when differentiating, and $\lambda_D = 0.05$, $\lambda_R = 0.02$. We use pessimistic value estimates from paired networks; the arguments (s, σ, a, z) of Q_D and Q_R are omitted for readability. The first term scores the latent task, Q_D pulls π_z toward the expert states that encode near z, and Q_R carries BFM-Zero's penalties. When the expert data contain few examples of a motion, the style prior may discourage it. Section IV-E tests whether this explains why some prompts stop; the objective alone does not imply it.

E. Prompts without a common offset

After training, we use BFM-Zero's weighted reward readout over states $s_1, \ldots, s_n$ from its released replay buffer, with one change: we subtract SeeBFM's mean embedding $\bar{B} = n^{-1} \sum_i B(s_i)$ on those states,

$$z_r = \text{proj}\Big(\textstyle\sum_i w_i\, r(s_i)\big(B(s_i) - \bar{B}\big)\Big), \tag{7}$$

where $w_i = \exp(10 r(s_i)) / \sum_j \exp(10 r(s_j))$. This weighted, centered readout is an empirical inference rule, and as an estimate of u_r under the training distribution ρ it is biased. Before projection, the uncentered sum splits exactly into two terms,

$$\sum_i w_i\, r(s_i) B(s_i) = \sum_i w_i\, r(s_i)\big(B(s_i) - \bar{B}\big) + \bar{r}\, \bar{B}, \tag{8}$$

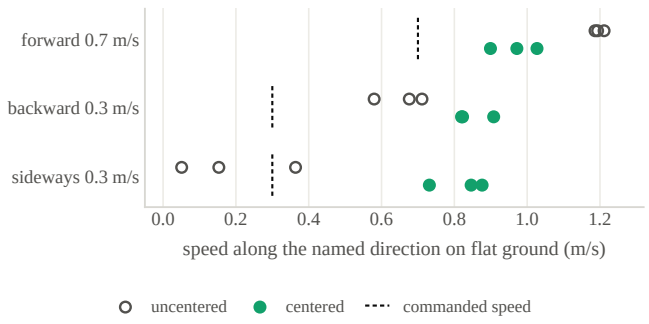


Fig. 3. Without centering, *sideways* 0.3 *m/s* barely moves sideways in two seeds: flat-ground speed along each prompt's named direction, with and without centering, one point per training seed.

with $\bar{r} = \sum_i w_i\, r(s_i)$. The second term lies along the same vector $\bar{B}$ for every reward, scaled by its weighted mean $\bar{r}$. The size of this offset depends on the inference states. Over SeeBFM's buffer, the mean of B has norm 1.0, compared with 16 for each normalized $B(s)$; over BFM-Zero's buffer, its norm is 4.4. A constant reward has a policy-independent return in a continuing task, but that does not force its embedding $\mathbb{E}_\rho[B]$ to be small. We therefore assess centering empirically: the uncentered latent of *forward* 0.7 *m/s* has cosine 0.86 with $\bar{B}$, the centered one 0.33, and centering restores the tested prompts' named directions (Fig. 3). BFM-Zero is prompted with the prompts released with it, which we call BFM-Zero's prompts, and SeeBFM with its own centered prompts for the same six locomotion rewards: *forward* 0.3 *m/s*, *forward* 0.7 *m/s*, *backward* 0.3 *m/s*, *sideways* 0.3 *m/s*, *forward, arms raised* and *turn in place*, the first four being the walking prompts, plus sixteen further rewards from its task set as a check that the results are not specific to six.

F. From a scan to depth cameras

The scan is privileged, so a deployed robot must build it from its own sensors. A depth student predicts the scan $\hat{\sigma}$ from two depth cameras and the robot's proprioception, and the frozen policy reads $\hat{\sigma}$ in the scan's place, so the policy, F, B and every prompt stay as they are (Fig. 4). SeeBFM, frozen, acts as its teacher with the true scan in simulation, half its robots tracking clips and half following reward prompts, while 20.5M frames of proprioception, scan, link poses and actions are recorded. Depth is rendered from those poses for Unitree's D435i on the head (106×60 pixels) and a ZED X Mini at the torso's lower front (96×60), placed as in [20], with the robot's own body in view, its sensor's noise, a camera error drawn per robot, latency and dropped frames. The depth student is first trained on

$$\mathcal{L}_\sigma = \frac{\sum_j m_j\, \ell_\delta(\hat{\sigma}_j - \sigma_j)}{\sum_j m_j}, \tag{9}$$

where j runs over the 598 rays, ℓ_δ is the Huber loss with $\delta = 5$ cm, and m_j is 1 if either camera saw ray j's ground in the last 2 s and 0.1 otherwise. This emphasizes recently visible ground while retaining supervision elsewhere. We then fine-tune it on $\mathcal{L}_\sigma + \lambda_a \|\mu(s, \hat{\sigma}, z) - a^*\|^2$ over the recorded rollouts with reward prompts, where μ is the frozen actor's mean action and a^* is the recorded teacher action. Gradients pass through the frozen actor to the student, so scan errors are also judged by their effect on control; λ_a makes the two terms equal at the start.

G. From words to prompts

We answer a typed command with a behavior that SeeBFM is known to carry out. Every flat-ground motion of BONES-SEED [2] in our training split becomes a behavior, prompted as BFM-Zero tracks a motion, with the latent of the next reference state at each step, $z_t = \mathrm{proj}(B(s^{\mathrm{ref}}_{t+1}))$. Walking drifts in heading, so a traveling behavior is steered in z: its prompt moves against the centered prompt of the pelvis's yaw rate, in proportion to the robot's turn from its starting heading. A judge counts a robot as doing a behavior if it does not fall and its motion, compared by dynamic time warping, is nearer the behavior's reference than any of 20 distractors. The library keeps the behaviors SeeBFM does on flat ground in at least half of its attempts and whose motion matches what their descriptions claim, 6,822 of 7,261.

The language step chooses from the library. A sentence encoder, all-MiniLM-L6-v2 [22, 34] fine-tuned so that two descriptions of one behavior match and descriptions that read alike but move differently do not, returns the 30 behaviors whose descriptions best match the command. A local language model, Qwen3-8B [39], reads them and picks one, or answers that none fits and the robot stands. The chosen behavior plays as a sequence of motion prompts, and the trained policy is again left as it is. The baseline, one latent per command, is an MLP trained on BONES-SEED's 1.38M pairs of a text and its clip to map the text's embedding to the centered mean of B over the clip's frames; its latent is held for the whole episode.

H. Data and training

The corpus holds 379.2 h of G1 motion from seven sources: BONES-SEED by Bones Studio [2], AMASS [19, 47, 21], GRAIL [37], PARC [38], OmniRetarget [40], EgoHTR [3] and LAFAN1 [8]. Of these, 77.7 h are on terrain such as stairs, boxes, seats and uneven ground, placed in one static atlas in the simulator, each clip on its own terrain. Clips are drawn in proportion to their length, with terrain clips raised from their 20.5% share to 40% of draws, and BFM-Zero's prioritization of poorly tracked clips raises the realized share to about 53%. We hold out 1% of clip groups per dataset for validation. The renders in Figs. 1, 2 and 4 show motion and terrain from LAFAN1, GRAIL, PARC and BONES-SEED.

The scan has 598 rays on two grids in the torso's yaw frame: 13×25 at 0.1 m from 0.8 m behind to 1.6 m ahead and 0.6 m to either side, and 13×21 at 0.05 m from 0.3 m behind to 0.7 m ahead and 0.3 m to either side. Each ray reads the torso's height above the surface below it. The scan is privileged and noise-free, leaving sensor noise to the depth student, and the encoder e is a small CNN per grid with 128 outputs.

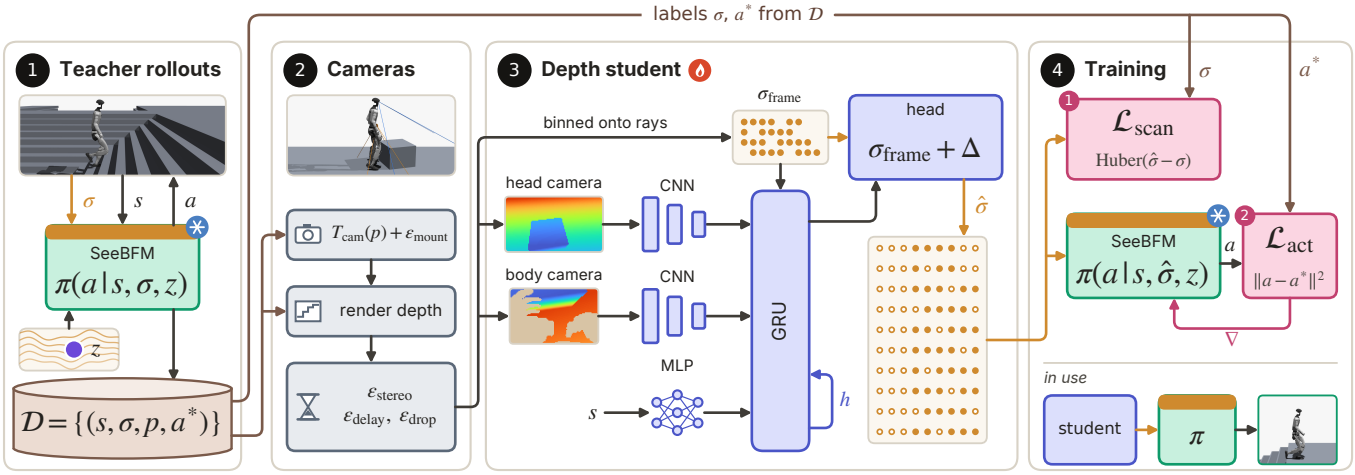


Fig. 4. **The depth student**. (1) Frozen SeeBFM acts as a teacher. (2) Each camera's depth is rendered. (3) A network per camera turns depth into a prediction of the scan. (4) The student learns the scan, then refines it so the frozen policy acts as it does with the true scan.

Training continues from the released checkpoint with BFM-Zero's settings (1,024 environments, 16 gradient updates per step of all environments, discount 0.98) and a fresh optimizer. The released weights train at 0.6 of their released learning rate, B at 0.3 and the new weights at the full rate, and between 19.2M and 28.8M steps every rate steps down to 0.05 of its value. Three training seeds run 28.8M steps each, about 13 h per seed on one H100. For typed commands, the language model restates each command literally before the search and reasons before it picks, and a second take of the same motion passes the judge 87 to 88% of the time.

I. Evaluation

Crossing is measured in Isaac Lab on a grid of stairs going up, stairs going down and boxes, with a terrain seed unseen in training. Each attempt starts on a 1.2 m platform surrounded by the terrain and succeeds if the robot reaches 1.7 m beyond the platform's edge within 20 s without a fall, a fall being contact between the ground and any body other than the feet for five consecutive control steps. We report the four rows with 0.10 to 0.15 m steps, about 230 attempts per terrain and seed with the six prompts pooled, and falls per 1,000 control steps over all rows. Where reported as a percentage, falls instead measure the share of attempts that fall on the stated rows. The evaluation keeps BFM-Zero's training perturbations: a push of 0.5 m/s every 1 to 3 s, observation noise, and randomized masses, friction and default joint angles. Table I names the models and conditions we compare. Two of them take perception away. The level scan shows whether the trained policy uses what it sees, and the model trained without the scan, the same recipe with every scan pathway held at zero, tests whether this training recipe learns the skill without the scan. In the text, the level scan, the seen-ground scan and the depth student stand for the SeeBFM conditions of those names, and a prompt's name is set in italics, as in *forward* 0.7 *m/s*.

To measure how a prompt's behavior changes, we compare its speed on terrain with its speed on flat ground. For a walking prompt z, $\bar{v}_T(z)$ is the robot's velocity along the prompt's direction in its own heading frame, averaged over every robot and control step on terrain T, or its yaw rate for *turn in place*, and the prompt keeps

$$\kappa_T(z) = \bar{v}_T(z) \,/\, \bar{v}_{\text{flat}}(z) \tag{10}$$

of its flat-ground speed on T, provided the denominator is nonzero. This measures the behavior a prompt produces; the reward it specifies stays the same. Tracking error is BFM-Zero's earth mover's distance on joint angles, over 1,769 flat and 439 terrain validation clips. That distance can stay small for a robot that never leaves the bottom step, so on the terrain clips we also count ascents: a clip whose reference pelvis rises by $\Delta^{\text{ref}} = \max_t h_t^{\text{ref}} - h_0^{\text{ref}} \geq 0.10$ m is climbed if the robot's

TABLE I
THE MODELS AND CONDITIONS COMPARED. EVERY SEEBFM CONDITION RUNS THE SAME TRAINED POLICY AND CHANGES ONLY WHAT IT READS IN PLACE OF THE TRUE SCAN.

Name	What it is
BFM-Zero	the released model, with the prompts released with it
SeeBFM	the trained model, reading the true height scan
SeeBFM, level scan	SeeBFM reading level ground at the height of the ground under its torso
SeeBFM, seen-ground scan	SeeBFM reading the true scan only where a camera saw the ground in the last 2 s
SeeBFM, depth student	SeeBFM reading the depth student's prediction from two cameras
trained without the scan	SeeBFM's recipe with every scan pathway held at zero
P-BFM, with command	the P-BFM policy its authors released, given the prompt's direction and speed as its command
P-BFM, prompt only	the same policy given the prompt alone

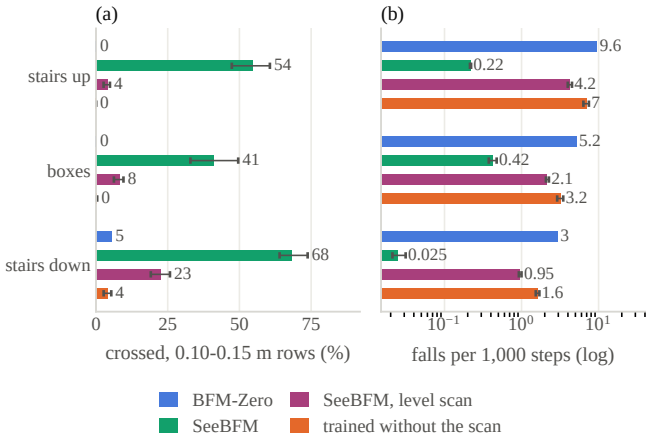


Fig. 5. Crossing drops when the scan is made level or omitted during training: (a) crossing and (b) falls per 1,000 steps.

pelvis rises by $\max_t h_t - h_0 \geq \Delta^{\text{ref}}/2$. Results are means over three training seeds with the range in brackets, unless stated.

IV. Results

A. Flat-ground tracking and prompt behavior

We first check which behaviors remain after continued pretraining. Training with terrain improves tracking on flat ground, while most tested walking prompts retain their direction. Tracking error on flat validation clips falls from 0.93 at the start to 0.67 to 0.68 in the three seeds, and on terrain clips from 1.39 to about 1.12. With centered prompts, *forward* 0.7 *m/s*, *backward* 0.3 *m/s* and *sideways* 0.3 *m/s* move in their named directions on flat ground in every seed, and like BFM-Zero's they run faster than commanded, *forward* 0.7 *m/s* at a median of 0.89 m/s against BFM-Zero's 1.31. The exception is *forward, arms raised*: its reward per step on flat ground falls below BFM-Zero's in every seed, and in one seed it barely travels. Goal poses are a second exception, ending about 2 cm further from their targets than BFM-Zero's (Table IV). The goal poses are frames of LAFAN1, which was BFM-Zero's only training data and is 2.4 of SeeBFM's 379 hours, and the model trained without the scan misses them by more still. Thus the larger goal error is not specific to adding the scan; the changed data and training recipe may contribute.

B. Reward-prompted crossing depends on the scan

Under the training perturbations, with the same six locomotion prompts pooled, SeeBFM crosses stairs going up in 54.4% [47.4–60.6] of attempts, boxes in 41.0% [32.9–49.6] and stairs going down in 68.0% [64.0–73.8], where BFM-Zero crosses 0, 0 and 5.3% (Fig. 5), and it falls about twenty-five times less often. The pooled numbers average over prompts that behave very differently. The two forward prompts cross nearly every flight of stairs going up, *forward, arms raised* three in four and *sideways* 0.3 *m/s* more than half, while *backward* 0.3 *m/s* never crosses and *turn in place*, by design, stays where it is. The same prompt changes its gait with the ground: *forward* 0.7 *m/s* slows on stairs and boxes and lifts the swing foot over each step (Fig. 6). The skill holds beyond the six prompts: the sixteen further prompts cross 29.8% of stairs going up and 18.1% of boxes, where BFM-Zero crosses none.

With the level scan, SeeBFM crosses 4.0% of stairs going up and 8.1% of boxes with the same prompts, and on stairs going up every attempt that does not cross ends in a fall, so the trained policy relies on its scan. The three models trained without the scan, which start from the same weights and see the same data and seeds, cross no stairs going up and at most 0.4% of boxes, and fall far more often than SeeBFM (Fig. 5b). They are different policies from BFM-Zero, with their own failures and flat-ground speeds, yet they cross as rarely as it does. Within this training recipe, height-scan input was needed to learn the observed crossing skill.

The crossings above keep BFM-Zero's training perturbations, and with them off the picture changes in one respect. *Forward* 0.7 *m/s* still crosses, but *forward* 0.3 *m/s* halts at stair edges, so pooled crossing of stairs going up drops to 27.6% while falls almost vanish (0.6%), the rest of the attempts stopping short. Restored one at a time, the pushes alone carry *forward* 0.3 *m/s* past the edges, and they also cause most of the falls on stairs going up.

C. How prompts behave on terrain

We compare each prompt's speed along its direction on terrain with the same model's speed on flat ground (Fig. 7). Walking forward and turning carry over: averaged over stairs and boxes, *forward* 0.7 *m/s* keeps about three quarters of its flat-ground speed and *turn in place* most of its yaw rate, with a different gait. Backward and sideways walking slow substantially. On terrain *backward* 0.3 *m/s* keeps about a tenth of its flat-ground speed and *sideways* 0.3 *m/s* about two fifths, and on boxes *backward* 0.3 *m/s* stops short in half of its attempts (Fig. 8).

These speed reductions occur in the models trained with the scan. The model trained without the scan keeps three quarters of its backward speed and more than half of its sideways speed on the same terrain (Fig. 7, orange), and falls: walking backward over boxes, every one of its attempts ends on the ground. The backward stop also survives with the training perturbations off, where *backward* 0.3 *m/s* keeps about a twentieth of its flat-ground speed on stairs (Fig. 7, hollow dots). The robot sees the ground it would walk onto, since the scan reaches 0.8 m behind it and 0.6 m to either side, beyond where its feet land in the next half second at these speeds.

The motion data suggest one possible explanation. Of the corpus's moving time on terrain, 86.0% goes forward, 3.4% backward and 10.5% sideways, which leaves 2.0 h of backward and 6.1 h of sideways walking on terrain, against 51.1%, 16.6% and 32.3% on flat ground. Tracking such motion does not carry the robot up stairs either: scored by the rise of its torso, SeeBFM climbs none of 96 backward and 1 of 93 sideways stair ascents from clips built for the purpose, against 10 of 70 recorded forward ascents. A style prior trained on these data has seen few backward steps onto terrain, and Section IV-E tests whether that is why the prompts stop.

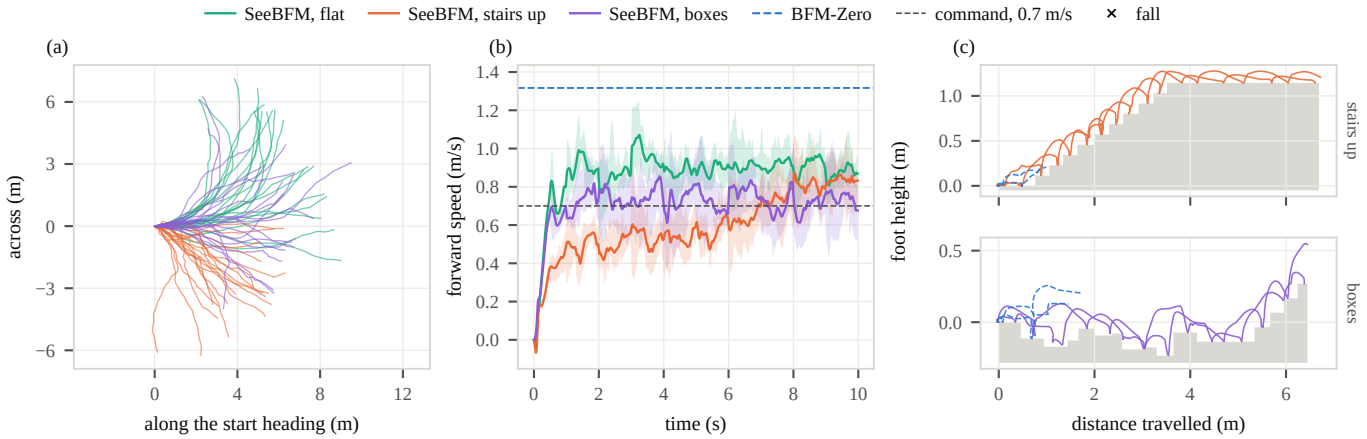


Fig. 6. (a) Paths, (b) forward speed and (c) swing-foot height over the terrain in one crossing, under the prompt *forward* 0.7 *m/s*.

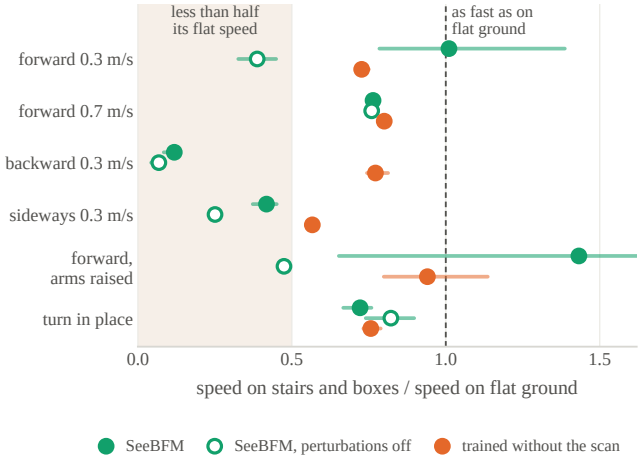


Fig. 7. Forward walking and turning retain much of their flat-ground speed on stairs and boxes; backward and sideways walking slow substantially.

D. Terrain tracking

Terrain motion stays harder to track than flat motion. Both improve by about 0.26 in tracking error over training, so the gap between them stays near 0.45: terrain tracking changes little after about 10M steps while flat tracking keeps improving. The three models trained without the scan keep the same gap (0.444 to 0.478, against 0.433 to 0.469). We found no evidence that sampling frequency explains this gap in the settings tested. Over training the discriminator draws each training frame about 7 times, against about 7,900 times for BFM-Zero's 2.4 h of LAFAN1, yet clips held out of training track as well as the training clips, and in shorter runs, changing the sampler, the learning rate, the number of updates or the capacity of B did not narrow the gap.

Flagged motion artifacts do not explain the gap either. A kinematic audit of the corpus flags 86 of the 439 terrain clips as physically implausible, with feet floating or sinking into the surface, weight resting on other body parts, or the body without support. SeeBFM tracks these clips better than the rest, and the gap widens slightly without them, so these flagged clips do not account for it. The worst-tracked clips are mostly PARC's short parkour clips, which pass the audit because a physics controller produced them for a human body, and whether the G1 can follow them remains open.

Counted by height, the scan does help terrain tracking. The three SeeBFM seeds climb 23.7 to 26.5% of the 257 ascents among the 415 terrain clips used for the ascent test, the three models trained without the scan 16.3 to 18.7%, and BFM-Zero 15.2%. Pooled over the three seeds, SeeBFM and the models trained without the scan climb 25.4 and 17.5%, respectively. Every model still fails most ascents. With one seed of each, the model trained without the scan has a tracking error close to SeeBFM's (1.19 against 1.10) and far below BFM-Zero's (1.51). Lower joint-angle error alone does not establish successful terrain traversal. The released checkpoints of Perceptive BFM [36] climb 4 to 7% of the same ascents. They were trained on a single motion, and the one that sees the terrain adapts a flat-ground version of each motion to the ground it sees, which clips captured on terrain cannot provide, so a fair comparison would retrain that tracker on our data.

Letting B read the scan made terrain tracking worse. With a scan encoder added to B's input or to its hidden layer, terrain tracking error at 9.6M steps was 1.31 against 1.14 for the same seed with $B(s)$, and the fit term of the forward-backward loss, a temporal-difference training diagnostic, rose by 12 to 30%. Bobrin et al. [1] saw the same degradation when B read their inferred contexts.

E. Testing the style-prior explanation

One hypothesis is that scarce backward steps onto terrain lead the style prior to discourage them. We first probe whether the trained discriminator assigns lower rewards to these rollouts. Along rollouts of each walking prompt, its reward drops from flat ground to terrain by 3.7 flat-ground standard deviations for backward walking and 2.4 for sideways walking, against 1.6 for forward walking. The drop comes mostly through the body's states: a level scan gives back only

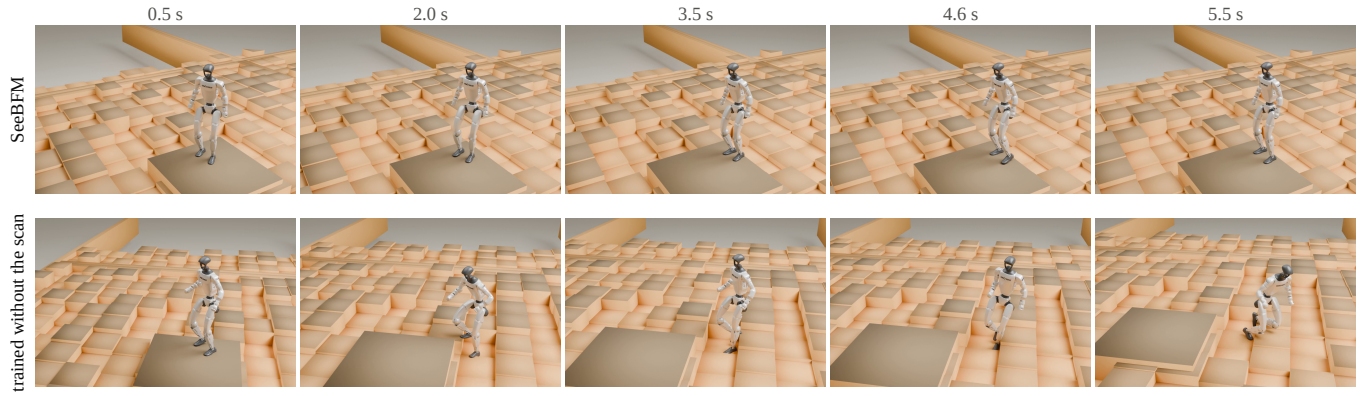


Fig. 8. *Backward* 0.3 *m/s* from the platform of a box cell: SeeBFM stops at the edge, and the model trained without the scan steps off and falls.

a quarter of each drop, and BFM-Zero's discriminator, which cannot see the terrain, also drops on its own backward and sideways rollouts on terrain (2.4 and 1.8). BFM-Zero walks backward on terrain anyway because it never trained there, so this verdict never shaped its policy, while SeeBFM trained on terrain under it. Only the discriminator's critic, which the actor maximizes, reads the scan strongly: a level scan gives back two thirds or more of its drop for every prompt.

The probe cannot separate cause from effect, so we added the missing motion. Flat clips of backward and sideways walking were moved onto stairs and boxes, each foothold placed on the surface and the joints solved by inverse kinematics. A candidate was kept if SeeBFM, tracking it on its own terrain, neither fell nor tracked it worse than the median recorded terrain clip; 18,256 passed (33.6 h), and 30.2 h of them were added to the corpus and to the discriminator's expert data. Backward walking then makes up 15.3% of the moving time on terrain and sideways walking 22.8%. Two training runs on the completed corpus track held-out added clips with lower error (0.77 against 0.83), but the prompts' speeds on terrain barely change: backward walking gains speed mainly going down stairs, and sideways walking slows on stairs and boxes and keeps its speed on flat ground. Adding these examples did not recover the requested behavior, so this intervention does not establish data scarcity as the cause. Whether the stop reflects a learned preference or a missing control capability stays open. A generator of terrain motion conditioned on the scan and a direction, improved with SeeBFM's tracking as its reward as Olkin et al. [20] do for their own generator, would let the model extend its own corpus.

F. Which states to infer prompts over

Forward-backward theory uses the training reference distribution ρ [30]. Our weighted readout can use a different sample set, so we compare inference buffers and centering directly. Inferred over SeeBFM's buffer, sampled as training samples it, uncentered prompts reach their named directions, yet they cross no more often than BFM-Zero's prompts and overshoot their commanded speed by about 0.5 m/s; centered, no seed reaches BFM-Zero's flat-ground reward on four of the six prompts (Table II). Over the motion corpus, where $\bar{B}$ is larger still (norm about 6), uncentered prompts miss their directions and centering corrects them, but they cross only 29 to 31% of stairs and boxes. Centered prompts over BFM-Zero's buffer have the highest crossing rate and meet the flat-ground reward criterion in all three seeds. Unless specified, we use them for SeeBFM's reward prompts. The skill does not require this readout, however: given BFM-Zero's prompts, SeeBFM still crosses 48.1% of stairs going up and 26.4% of boxes.

TABLE II

REWARD PROMPTS BY THE STATES THEY ARE INFERRED OVER, THREE SEEDS. DIRECTION: SPEED ALONG THE NAMED DIRECTION OVER THE COMMAND, LOWEST SEED. CROSSING: STAIRS UP AND BOXES, SIX / SIXTEEN PROMPTS. REWARD: SEEDS THAT REACH BFM-ZERO'S FLAT-GROUND REWARD ON FOUR OF SIX PROMPTS. SPEED: ERROR OF THE WALKING PROMPTS' SPEED.

States	Centered	Direction	Crossing (%)	Reward	Speed (m/s)
BFM-Zero's prompts		1.00	37.2 / 16.1	3	0.28
BFM-Zero's buffer	no	0.72	34.1 / 18.7	1	0.29
BFM-Zero's buffer	yes	1.00	47.7 / 23.9	3	0.36
SeeBFM's buffer	no	1.00	36.7 / 18.9	3	0.52
SeeBFM's buffer	yes	1.00	37.8 / 15.9	0	0.58

G. Seeing through depth cameras

The depth student keeps most of SeeBFM's skill (Fig. 9). It crosses stairs nearly as often as SeeBFM, boxes less often, and falls more on every terrain. The visibility control suggests that unseen ground accounts for much of this difference (Fig. 10): where the current frame measures the ground, the depth student's scan follows it closely, and behind the robot, on ground neither camera has seen, it is off by 6 to 10 cm on average over the attempt. With the seen-ground scan, the true scan kept only where a camera saw the ground in the last 2 s, SeeBFM falls only a few points less often than with the depth student on every terrain. This is a visibility control; it does not bound what a learned predictor could infer about unseen ground. Fine-tuning through the frozen actor lowers falls in

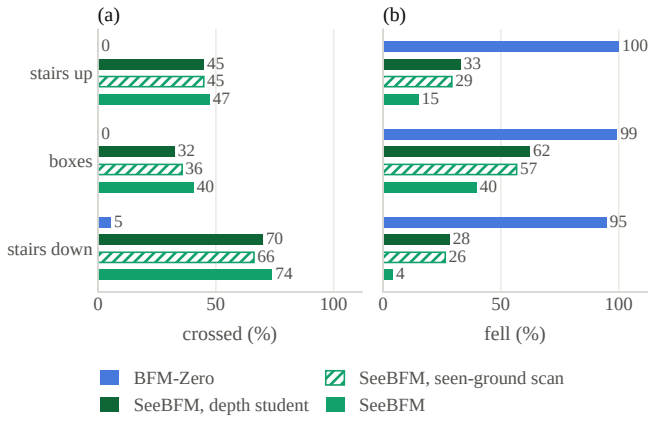


Fig. 9. (a) Crossing and (b) falls, in % of attempts on the 0.10–0.15 m rows.

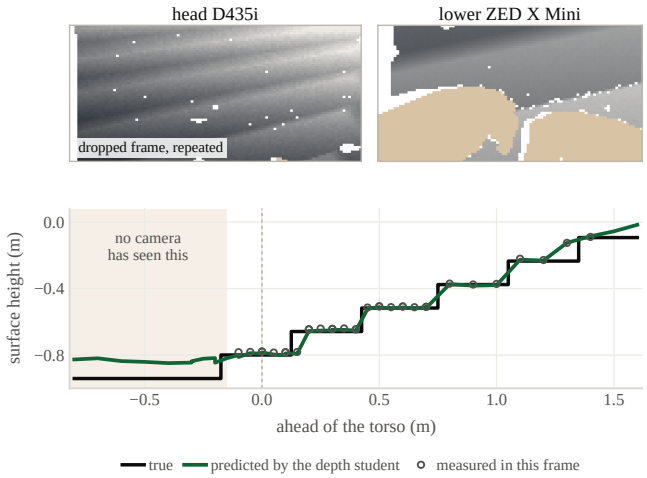


Fig. 10. The depth student 1.14 s into a stair climb. Top: the two cameras' depth as it receives them, darker nearer, white without a return, tan where its own legs block the view. Bottom: the surface height ahead of the torso, true, predicted and measured in this frame.

every training seed, with nearly all of the gain on stairs going down. DAgger on the depth student's own rollouts [23] gave a gain of similar size, but the reported results do not establish a consistent advantage. Stairs going down remain the depth student's limitation, though most of its falls there come after it has crossed: it gets down the measured flight without a fall in 94.0% of attempts, against SeeBFM's 77.3%, which stops short more often. On flat ground its six prompts match SeeBFM's to within 0.014 in reward per step.

H. Typed commands

We wrote 125 everyday requests, from walking, gestures and postures to dances, sports and requests no robot can do, and 12 that name a side. Each set was fixed before the runs that use it, and we chose the language step's settings on them. These sets therefore measure performance on development requests; an independent held-out language test remains to be done. We read each choice before any robot runs and count it right if the behavior does what is asked, with any side,

TABLE III
THE LANGUAGE STEP PASSES SEVEN OF THE ELEVEN COMMANDS AND ONE LATENT PER COMMAND PASSES TWO: EACH COMMAND'S CHECK ON FLAT GROUND, AND HOW MANY OF ITS THREE WORDINGS PASS IT. WALKING MUST ALSO STAY WITHIN 45° OF ITS STARTING HEADING; THE WALK CONTROL IS *forward* 0.7 *m/s*.

Command	Check	One latent per command	Language step
walk forward	forward ≥ 0.3 m/s	0	3
walk forward slowly	forward 0.05 to 0.6 m/s, below the walk control	0	2
jog forward	forward at least the walk control's $+0.2$ m/s	0	0
walk backward	forward ≤ -0.15 m/s	1	3
side step left	leftward ≥ 0.1 m/s and above forward	0	3
turn around in place	a net turn of 135 to 225°, never past 225°, at most 30° in the last 4 s, speed ≤ 0.6 m/s	0	0
wave the right hand	right wrist above the shoulder in $\geq 50\%$ of steps, waving at ≥ 0.5 Hz, left hand up in $\leq 20\%$	0	0
walk forward while waving	forward ≥ 0.2 m/s, wrist up in $\geq 15\%$, waving as above	0	3
crouch down	mean pelvis height ≤ 0.60 m	2	3
jump in place	pelvis peak ≥ 0.08 m above its median, speed ≤ 0.4 m/s	0	0
stand still	speed ≤ 0.1 m/s, yaw rate ≤ 0.15 rad/s	2	2

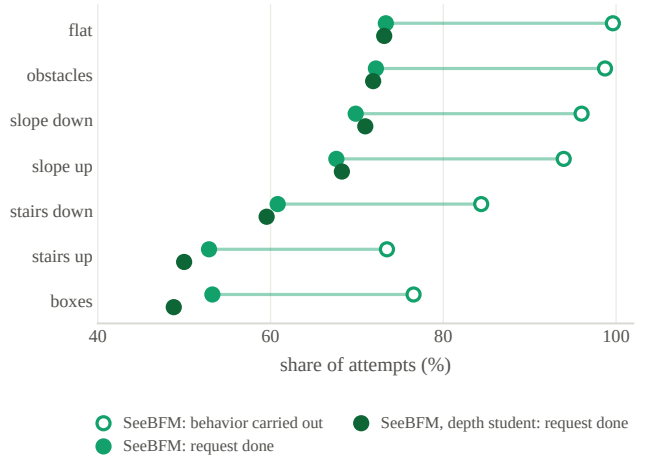


Fig. 11. A wrong choice loses about a quarter of the everyday requests.

direction or speed stated. The language step picks a right behavior for 92 of the everyday requests, of the 111 that have one in the library, and for 10 of the 12 that name a side.

Eleven commands, from walking to waving and jumping, each have a check on flat ground (Table III), and five reward prompts run beside every test as controls. Each command has

TABLE IV
GOAL PROMPTS ON THE 0.10–0.15 M ROWS: MEDIAN BODY ERROR AFTER 20 S (MM) / FALLS (%), 21 GOAL POSES. THREE TRAINING SEEDS, EXCEPT FOR THE LEVEL-SCAN CONDITION, WHICH USES ONE.

	Flat	Stairs up	Stairs down	Boxes
SeeBFM	79 / 24	115 / 30	103 / 25	106 / 36
SeeBFM, level scan	79 / 23	91 / 68	106 / 42	102 / 60
Trained without the scan	90 / 25	103 / 71	113 / 56	108 / 56
BFM-Zero	59 / 24	78 / 77	90 / 60	87 / 67

three wordings, written before the language step was built, and passes if two of them do. The language step passes seven commands on two evaluation seeds and with the depth student, and one latent per command passes two. Jogging, turning around in place, the right-hand wave and jumping in place fail with both. The library's turns keep turning or add a walk, its greetings stay below the shoulder, and its jumps travel.

We then run every everyday request on each of Isaac Lab's seven terrain types, at the sizes of the crossing evaluation's rows and with most robots starting on the terrain (Fig. 11). SeeBFM does an everyday request end to end, chosen right and carried out, in 73% of attempts on flat ground and 53% on stairs going up and on boxes, and the depth student stays within 5 points of it on every type. The choice does not see the ground, so it costs the same on every type, and on flat ground it is the whole loss, since the chosen behavior is carried out in 99.6% of attempts. On stairs and boxes the robot also carries out fewer of the chosen behaviors, 73% on stairs going up. Requests naming a side are completed in 70–84% of attempts.

I. Goal prompts on terrain

A goal prompt encodes a target pose s^*, using the same mean embedding as the reward prompts: $\mathrm{proj}(B(s^*) - \bar{B})$. We prompt the 21 goal poses of BFM-Zero's code on flat cells and on the stairs and box rows, and measure how far the robot's bodies end from their places in the goal pose, relative to the pelvis (Table IV). A rule fixed before the rollouts asked that on terrain this error stay within 20 mm of flat ground and the share of falls within 10 points, and the goal prompts miss it: the error rises by 20 to 40 mm on every terrain type in every seed and the falls on boxes by 11 to 13 points, while BFM-Zero's error rises about as much from its lower base. The scan still keeps the goal poses upright: on terrain the level scan, the model trained without the scan and BFM-Zero all fall far more often than SeeBFM, while on flat ground all four fall in about a quarter of the rollouts, since some poses lie on the ground.

J. A concurrent perceptive model

P-BFM [35], concurrent with this work, trains a perceptive behavior model from scratch and steers it with a heading and velocity command beside the prompt. We scored the P-BFM policy its authors released, trained for 364.8M steps, in its own simulator, MJLab, on our rows with our crossing and fall rules, and SeeBFM in Isaac Lab, both with the training perturbations off (Table V). P-BFM's reward prompts are inferred with its own backward network over states from our reproduction of its training, since the authors did not release their replay. This comparison uses different simulators, sensors and training histories, so it does not isolate an architectural advantage.

Forward crossing rates are similar on these rows, including with the depth student. They also fall about as often within 2 s of a 1.0 m/s push, every prompt pooled over every row: on stairs going up, boxes and stairs going down, 8.1, 5.3 and 0.5% for SeeBFM and 11.0, 6.5 and 0.3% for P-BFM with its command. The models differ in how they are steered: with the prompt only, P-BFM's *forward* 0.7 *m/s* crosses 3.5% of stairs going up, and the authors' own forward preset 0% against 96% with the command, while SeeBFM's prompt alone crosses every flight. SeeBFM reaches this after 28.8M steps of continued training from BFM-Zero, against 364.8M steps from scratch. The training count excludes BFM-Zero's pretraining, so these counts do not measure total training efficiency.

Gaps separate the two models (Fig. 13). On platforms ringed by a trench 1 m deep whose width grows with the row, every model crosses the narrow trenches. Over wider ones P-BFM falls in most attempts with or without its command, while SeeBFM and the depth student cross more than twice as often and at the widest trenches often stop at the edge instead. Going slowly, SeeBFM stops in front of almost every trench of 28 cm or more, where P-BFM walks in. The trench was built after SeeBFM was trained, and P-BFM's released training configuration holds no gaps, though the terrain of the authors' own run is unknown to us.

The two also differ in prompt execution on terrain (Fig. 12). With its command, P-BFM's *backward* 0.3 *m/s* keeps its flat-ground speed on terrain and its *sideways* 0.3 *m/s* more than half of it, and both cross stairs more often than SeeBFM's at the price of many more falls, while SeeBFM's stop short. Stopping has a cost: SeeBFM also gives up crossings that P-BFM makes without a fall, such as *forward* 0.3 *m/s* going down stairs, 6% for SeeBFM against 75% for P-BFM. The sensors differ as well: SeeBFM's scan reads 0.8 m behind the robot, while P-BFM's camera and the depth student's two face forward, so walking backward or sideways both camera models move onto ground they have not seen. With its command, P-BFM covers more directions than the depth student, and the two fall about as often over the walking prompts. P-BFM's command does not restore SeeBFM's backward and sideways prompts either: added through projections that start at zero and fine-tuned for one seed, it held the commanded heading, cutting the walking prompts' yaw rate on flat ground to about a third, but raised backward and sideways crossing of stairs and boxes by only 5 to 6 points. We did not retain this variant. A one-GPU reproduction of P-BFM's recipe, modified to fit our compute, fell well short of the released policy, which is why we report the released one.

V. FUTURE WORK

Two experiments come next.

- *A generator-tracker baseline.* A published motion generator and tracker [20, 46] on our rows, commanded to

TABLE V

SEEBFM AND THE AUTHORS' P-BFM POLICY ON THE 0.10–0.15 M ROWS, EACH IN ITS OWN SIMULATOR WITH THE TRAINING PERTURBATIONS OFF, IN % OF ATTEMPTS. WALKING POOLS THE FOUR WALKING PROMPTS.

	Senses the terrain by	Forward 0.7 m/s, crossed			Walking, crossed			Walking, fell		
		up	boxes	down	up	boxes	down	up	boxes	down
SeeBFM	a height scan	100.0	81.2	97.0	33.6	35.6	38.5	0.9	13.5	1.4
SeeBFM, depth student	two depth cameras	98.3	85.4	97.2	43.7	35.0	43.0	24.3	35.0	16.3
P-BFM, prompt only	a depth camera	3.5	92.4	99.3	2.5	42.7	62.7	10.7	15.9	6.0
P-BFM, with command	a depth camera	97.2	84.7	93.1	38.6	39.5	71.4	27.5	41.7	15.3

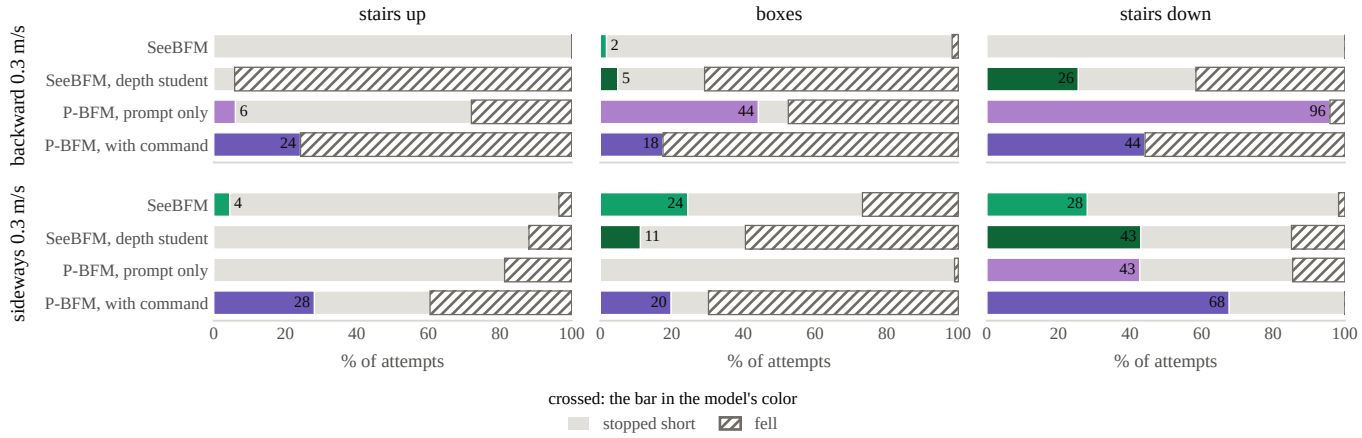


Fig. 12. Backward and sideways walking on the 0.10–0.15 m rows with the training perturbations off: crossed, stopped short and fell, in % of attempts.

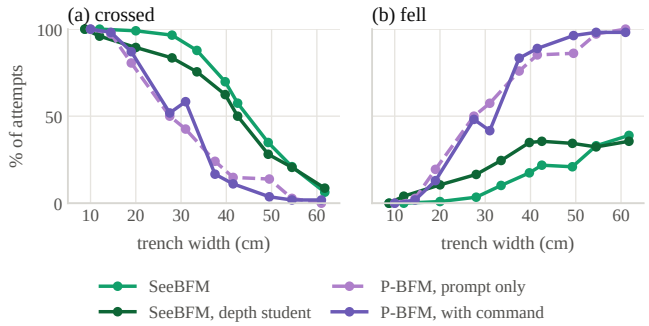


Fig. 13. *Forward* 0.7 *m/s* over a trench 1 m deep, by its width, with the training perturbations off: (a) crossed and (b) fell, in % of attempts; the rest stop short.

walk forward at the speeds of our forward prompts.

- *Hardware.* The depth student on a G1, on flat ground first and then on steps, with the walking prompts.

VI. LIMITATIONS

All results are in simulation. SeeBFM is trained and evaluated in Isaac Lab with a privileged, noise-free scan. The depth student is also evaluated in simulation only, and the P-BFM comparison runs each model in its own simulator. Tests on a G1 with the depth student are in preparation. Crossing varies across our three training seeds, from 47.4 to 60.6% on stairs going up. Pooled stair crossing is lower without training perturbations; the slow forward prompt relies on pushes to pass stair edges. *Backward* 0.3 *m/s* nearly stops on terrain and *sideways* 0.3 *m/s* slows by more than half. The heading of *forward* 0.7 *m/s* drifts on every terrain. Tracking prompts climb about a quarter of the terrain clips' ascents. Typed commands are done end to end on stairs and boxes a little over half the time, 14 of the 125 everyday requests have no behavior in the library, and turning around in place and jumping in place fail. The request sets also informed the language step's settings. Goal poses end further from their targets than BFM-Zero's even on flat ground, both with and without scan training, and further still on terrain.

VII. CONCLUSION

We add terrain perception to a pretrained behavior foundation model while retaining reward, goal and motion prompting. Placed on the forward side of the factorization and trained from the released weights, a height scan supports terrain crossing that the same recipe does not learn without it, alongside better tracking on flat ground. A depth student supplies the scan from two cameras without retraining the policy, and typed commands select motions for that policy. These results are in simulation; deployment on a G1 remains future work. Perception also changes prompt execution: some behaviors carry over to terrain, while others slow or stop. The scan controls help distinguish these responses from failures caused by missing perception. This is why we evaluate a perceptive behavior model prompt by prompt, beyond pooled success rates.

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